

RankUp NEET — Ultra Premium Sheet

Electrostatics • Target: AIR <100 • 20 Questions

Q1. Two charges q and $2q$ are placed at distance r . Force becomes zero at point where distance from q is:

- A. $r/3$
- B. $r/2$
- C. $r/\sqrt{2}$
- D. r

Q2. Electric field inside conductor is:

- A. max
- B. zero
- C. constant
- D. variable

Q3. Potential due to point charge varies as:

- A. $1/r^2$
- B. $1/r$
- C. r
- D. constant

Q4. (Assertion-Reason)

A: Electric field inside hollow conductor is zero.

R: Charges reside on surface.

- A. Both correct explain
- B. Both correct not explain
- C. A correct R wrong
- D. Both wrong

Q5. Capacitance depends on:

- A. charge
- B. potential
- C. geometry
- D. current

Q6. Force between charges doubles if distance becomes:

- A. $r/2$
- B. $2r$
- C. r^2
- D. $\sqrt{r}$

Q7. Work done moving charge in equipotential surface:

- A. zero
- B. positive
- C. negative
- D. infinite

Q8. Electric field lines never:

- A. start
- B. end
- C. intersect
- D. curve

Q9. Dipole moment unit:

- A. C

- B. Cm
- C. Nm
- D. J

Q10. Field at center of dipole:

- A. zero
- B. max
- C. infinite
- D. constant

Q11. Potential gradient gives:

- A. field
- B. force
- C. charge
- D. energy

Q12. Parallel plate capacitor energy:

- A. CV^2
- B. $\frac{1}{2}CV^2$
- C. QV
- D. none

Q13. Dielectric increases:

- A. field
- B. capacitance
- C. charge
- D. resistance

Q14. Electric flux unit:

- A. Nm^2/C
- B. N/C
- C. C
- D. J

Q15. Gauss law valid for:

- A. symmetric cases
- B. all cases
- C. only sphere
- D. none

Q16. Potential inside conductor:

- A. zero
- B. constant
- C. variable
- D. max

Q17. Field due to infinite sheet:

- A. varies
- B. constant
- C. zero
- D. infinite

Q18. Capacitance of isolated sphere depends on:

- A. charge
- B. radius
- C. potential
- D. none

Q19. Energy density in electric field:

A. $\frac{1}{2}\epsilon_0 E^2$

B. $\epsilon_0 E^2$

C. E^2

D. none

Q20. Work done bringing charge from infinity:

A. qV

B. V/q

C. zero

D. infinite

Answer Key

Q1: A
Q2: B
Q3: B
Q4: A
Q5: C
Q6: A
Q7: A
Q8: C
Q9: B
Q10: A
Q11: A
Q12: B
Q13: B
Q14: A
Q15: B
Q16: B
Q17: B
Q18: B
Q19: A
Q20: A